

Disproof of Conjecture 00000001192

The stated leading coefficient is not an integer

1 The conjecture

Conjecture 1 (as filed). *Let $c(n, q)$ denote the number of subgroups of $\mathrm{GL}_n(\mathbb{F}_q)$ whose order equals that of the commutator (derived) subgroup. Then $c(n, q)$ is an integer-coefficient polynomial in q of degree $n^2 - n$ with leading coefficient*

$$\mathrm{LC}(n) = \frac{\prod_{p|n} p}{n! n^n}. \quad (1)$$

Verdict: FALSE. The two structural assertions in Conjecture 1 — that the polynomial has *integer coefficients* and that its leading coefficient is $\mathrm{LC}(n)$ as in (1) — are mutually inconsistent. We make no claim about the actual value of $c(n, q)$; the refutation concerns only the stated kind of polynomial and the stated leading coefficient.

2 The claimed leading coefficient

Reducing (1) to lowest terms, write $\mathrm{LC}(n) = a_n/b_n$. Since $\mathrm{rad}(n) := \prod_{p|n} p$ divides n , and $n \mid n^n \mid n! n^n$, we always have $\mathrm{rad}(n) \mid n! n^n$; hence $a_n = 1$ and $b_n = n! n^n / \mathrm{rad}(n)$ for every $n \geq 1$ (with $\mathrm{rad}(1) = 1$). The first values are recorded in Table 1.

n	degree $d = n^2 - n$	$\mathrm{LC}(n)$	integer?
1	0	1	yes
2	2	1/4	no
3	6	1/54	no
4	12	1/3072	no
5	20	1/75000	no
6	30	1/5598720	no

Table 1: The claimed leading coefficient $\mathrm{LC}(n)$, exactly. For every $n \geq 2$ the reduced numerator is 1 and the reduced denominator exceeds 1, so $\mathrm{LC}(n)$ is a non-integer rational. The degree $d = n^2 - n \geq 2$ for all $n \geq 2$.

For instance,

$$\mathrm{LC}(2) = \frac{2}{2! \cdot 2^2} = \frac{2}{8} = \frac{1}{4}, \quad \mathrm{LC}(3) = \frac{3}{3! \cdot 3^3} = \frac{3}{162} = \frac{1}{54}, \quad \mathrm{LC}(4) = \frac{2}{4! \cdot 4^4} = \frac{2}{6144} = \frac{1}{3072}.$$

Proposition 1. *For every integer $n \geq 2$, $\mathrm{LC}(n)$ is not an integer.*

Proof. As noted, $\text{rad}(n) \mid n$, so $\text{rad}(n) \mid n^n$, and therefore $\text{rad}(n) \mid n!n^n$. Thus the reduced numerator of $\text{LC}(n) = \text{rad}(n)/(n!n^n)$ is 1, while the reduced denominator is $n!n^n/\text{rad}(n)$. For $n \geq 2$ we have $\text{rad}(n) \leq n$, hence

$$\frac{n!n^n}{\text{rad}(n)} \geq \frac{n!n^n}{n} = n!n^{n-1} \geq 2 \cdot 1 = 2 > 1,$$

using $n! \geq 2$ and $n^{n-1} \geq 1$. A rational with reduced numerator 1 and reduced denominator > 1 is not an integer. $\square$

3 Why this refutes the conjecture

3.1 The integer-coefficient reading

Suppose, as Conjecture 1 asserts, that $c(n, q)$ is a polynomial

$$c(n, q) = a_d q^d + a_{d-1} q^{d-1} + \cdots + a_0, \quad a_j \in \mathbb{Z},$$

of degree $d = n^2 - n$. The leading coefficient a_d is then an integer, because it is one of the integer coefficients of the polynomial. But the conjecture also identifies a_d with $\text{LC}(n)$. Taking $n = 2$ (so $d = 2$ and $\text{LC}(2) = 1/4$), this would require the integer a_2 to equal $1/4$, i.e. there would have to exist $a \in \mathbb{Z}$ with

$$4a = 1.$$

No such integer exists. The same contradiction occurs at $n = 3$ ($a_6 = 1/54$), $n = 4$ ($a_{12} = 1/3072$), and in fact at every $n \geq 2$ by Proposition 1. Consequently no polynomial with integer coefficients of the stated degree has the stated leading coefficient: Conjecture 1 is false as written.

3.2 The charitable “integer-valued” reading

One might weaken “integer-coefficient” to “integer-valued” (i.e. $c(n, q) \in \mathbb{Z}$ for all integers q), since a PORC formula often only needs to take integer values. This does not rescue the conjecture. A classical fact about integer-valued polynomials is:

Proposition 2. *If $P \in \mathbb{Q}[x]$ is integer-valued and has degree d , then $d! \cdot (\text{leading coefficient of } P) \in \mathbb{Z}$.*

Sketch. The binomial polynomials $\binom{x}{k}$ ($k = 0, \dots, d$) form a $\mathbb{Z}$ -basis of the integer-valued polynomials of degree $\leq d$. In this basis $P = \sum_{k=0}^d b_k \binom{x}{k}$ with $b_k \in \mathbb{Z}$, and $\binom{x}{k}$ has leading coefficient $1/k!$. Hence the leading coefficient of P is $b_d/d!$, so $d!$ times it is $b_d \in \mathbb{Z}$. $\square$

Applying Proposition 2 to the conjectured degree $d = n^2 - n$:

- $n = 2$: $d = 2$ and $2! \text{LC}(2) = 2 \cdot \frac{1}{4} = \frac{1}{2} \notin \mathbb{Z}$;
- $n = 3$: $d = 6$ and $6! \text{LC}(3) = 720 \cdot \frac{1}{54} = \frac{720}{54} = \frac{40}{3} \notin \mathbb{Z}$.

So even under the weaker integer-valued reading, the claimed leading coefficient is impossible at the two smallest cases $n = 2$ and $n = 3$. The ambiguity between the two readings does not rescue the conjecture.

3.3 Arithmetic content, in divisibility form

The two obstructions used above are exactly the following elementary facts, which are the ones formalised in the accompanying Lean development:

$$\nexists m \in \mathbb{N} (m \cdot 4 = 1), \quad \nexists m \in \mathbb{N} (m \cdot 2 = 1), \quad \nexists a \in \mathbb{Z} (4a = 1),$$

and, for the integer-valued reading at $n = 3$,

$$\nexists m \in \mathbb{N} (m \cdot 54 = 720) \quad \left(\frac{720}{54} = \frac{40}{3} \right).$$

4 Scope, and the PORC caveat

The quantity $c(n, q)$ itself — the number of subgroups of $\mathrm{GL}_n(\mathbb{F}_q)$ whose order equals that of the derived subgroup — is not otherwise analysed in this submission. The refutation is entirely about the *stated leading coefficient* being arithmetically impossible for the *stated kind of polynomial* (integer-coefficient, degree $n^2 - n$). This is a self-contained inconsistency in the text of the conjecture, independent of any group theory.

If the author intended a PORC description, in which the coefficient is a constituent depending on $q = p^f$ or on the residue class of q , or intended a different normalisation (for example, a degree-0 or lower-degree polynomial, or a coefficient such as $\mathrm{rad}(n)/(n!n^n)$ viewed only after multiplication by a further factor), then the formula as written is still self-inconsistent: as written it demands an integer-coefficient polynomial whose leading coefficient is the non-integer rational $\mathrm{LC}(n)$. To make the statement consistent one must change at least one of: the class of polynomials (allowing rational coefficients), the degree, or the normalisation of the leading coefficient. No such qualification appears in the conjecture as filed.

Remark. Because the contradiction already occurs at $n = 2$ (degree 2), the refutation does not depend on any subtlety of Higman’s PORC phenomenon or on the actual count $c(n, q)$; it is a statement about rational arithmetic.

Reproducing

The exact values of $\mathrm{LC}(n)$ for $n = 1, \dots, 12$ (and the $d! \mathrm{LC}(n)$ integer-valued test) are produced by

```
python3 reproduce.py
```

which exits 0 on PASS. The finite arithmetic facts above are formalised in core Lean 4 under `lean4/` (no Mathlib, no `sorry`); build with `lake build` and audit with `lake env lean Check.lean`.